Weimin Qiu

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Google Scholar | Homepage | GitHub | LinkedIn

Research Summary

My research focuses on generative visual reasoning and 3D understanding, enabling machines to reason about visual environments and physical dynamics. My work spans video generation and learning-based 3D reconstruction. I have contributed original research and actively serve as a peer reviewer for leading AI venues.

Education

University of California, Merced — Merced, CA | Jan 2025 – Present
Ph.D. in Computer Science — Advisor: Meng Tang

University of California, Merced — Merced, CA | Sep 2023 – Dec 2024
M.Sc. in Computer Science — GPA: 3.86/4.0

Tongji University — Shanghai, China | Sep 2013 – Jun 2016
M.Eng. in Civil Engineering

Publications

1. **Weimin Qiu**, Jieke Wang, Meng Tang. "Self-Cross Diffusion Guidance for Image Synthesis of Similar Subjects." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2025. [Paper]
2. **Weimin Qiu**, Jieke Wang, Zhining Gu, Vincent Tao Hu, Meng Tang. "Diversifying Similar Subjects for Text-to-image Synthesis." *International Journal of Computer Vision (IJCV)*, July 2026. [Paper]
3. Zhenghao Ni, **Weimin Qiu**, Meng Tang. "Learning via Self-Consistency for Diffusion-based Video Reasoning." *Under review.* [Paper]
4. Zhining Gu, Shangjie Du, **Weimin Qiu**, Carl Olsson, Ping Liu, Meng Tang. "Relative Camera Pose Estimation via Learnable Ranker and Verifier." *Under review.* [Paper]

Research and Professional Experience

University of California, Merced — Merced, CA | Nov 2023 – Present
Research Assistant – Advisor: Meng Tang

- Developed Self-Cross Diffusion Guidance, penalizing overlap between cross-attention and aggregated self-attention maps; reduced subject mixing compared with methods relying on either signal alone. (CVPR 2025)
- Proposed Diversity Guidance using diffusion-model internal features for instance segmentation and improved subject-level diversity over prompt engineering; extended these internal signals into rewards for RL post-training. (IJCV 2026)
- Developing an on-policy distillation method motivated by internal consensus in foundation models to improve video diffusion models on 2D and 3D reasoning tasks.

University of California, Merced — Merced, CA | Sep 2024 – Present
Teaching Assistant – Full-Stack Development and Deep Learning

State Grid Corporation of China — Shanghai, China | Sep 2016 – Feb 2022
Engineer & Project Manager – Construction of 10–110 kV substations

Reviewer Service

Peer reviewer for International Conference on Learning Representations (ICLR), European Conference on Computer Vision (ECCV), IEEE Transactions on Image Processing (TIP), AAAI Conference on Artificial Intelligence (AAAI), Asian Conference on Computer Vision (ACCV), and Winter Conference on Applications of Computer Vision (WACV).